COMPA$_test$

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Robot kinematics

At each instant, the robot's velocity (or twist) $\mathbf{v}$ is related to the actuator rates ω through the Jacobian:

$$\mathbf{v} = J\,\omega.$$

Bounded actuators

Motors cannot spin infinitely fast, so we assume their rates are bounded:

$$\|\omega\| \leq 1.$$

This means all allowable ω lie inside a unit ball.

What twists are achievable?

The set of possible $\mathbf{v}$ is the image of that ball under J:

$$\mathcal{V} = \{\, J\,\omega : \|\omega\| \leq 1 \,\}.$$

A basic fact from linear algebra: the image of a sphere under a linear transformation is an ellipsoid.

Ellipsoid shape

- The ellipsoid's axes directions come from the left singular vectors of J.
- The ellipsoid's axes lengths are the singular values of J.

Thus, if one singular value is small, motion in that direction is hard (short axis).

Equation of the ellipsoid

To describe the ellipsoid mathematically, take any velocity $\mathbf{v}$. The minimum-norm actuator input that achieves it is:

$$\omega^* \;=\; J^\top (J J^\top)^{-1}\,\mathbf{v}.$$

Its squared norm is:

$$\|\omega^*\|^2 \;=\; \mathbf{v}^\top (JJ^\top)^{-1}\mathbf{v}.$$

If this is ≤ 1, the twist is feasible. Hence the *mobility ellipsoid* equation:

$$\mathbf{v}^\top (JJ^\top)^{-1}\mathbf{v} \;\leq\; 1.$$